

Part V — Waves, Sound, and Light · Chapter 25

Sound Waves and Resonance

How longitudinal mechanical waves create the experience of sound — from whispers to music to sonar

A bat emits a pulse of ultrasound and receives the echo 0.004 seconds later. From this, it calculates the distance, size, and velocity of an insect 70 centimeters away — in complete darkness, while flying. A concert hall is designed so that every seat hears the same blend of direct and reflected sound, with reverberation times carefully tuned to the genre of music it was built for. A physician presses an ultrasound probe to a patient's abdomen and watches a fetal heartbeat on a monitor. A choir hits a resonant frequency of a cathedral and the stone walls themselves vibrate. In each case, the same physical phenomenon is at work: sound. This chapter applies the general wave principles from Chapter 24 to the specific, rich, and practically important world of acoustics.

25.1 What Sound Is

Sound is a mechanical disturbance that propagates through a material medium by the transmission of energy via oscillations of the medium's particles. It requires a material medium — air, water, steel, rock — and cannot travel through a vacuum. This was confirmed dramatically when Robert Boyle in 1660 showed that a bell ringing inside an evacuated glass vessel becomes inaudible as the air is pumped out.

What humans perceive as sound is a very small subset of all acoustic waves. The human auditory system detects pressure oscillations in air with frequencies from roughly 20 Hz to 20 000 Hz (20 kHz). Below 20 Hz is infrasound, inaudible to humans but produced by earthquakes, large machinery, and some animals (elephants, whales). Above 20 kHz is ultrasound, used in medical imaging, sonar, and industrial cleaning.

Sound is ultimately the energy of a vibrating source (vocal cords, a speaker diaphragm, a vibrating string) transferred to the surrounding medium and propagated outward. When these pressure oscillations reach the eardrum, they cause it to vibrate at the same frequencies, initiating the complex process of hearing.

Key Point: In space, no one can hear you scream — this is physically correct. Space is essentially a vacuum, and sound cannot travel without a medium. Explosions in space

films are cinematic fiction. Spacecraft can transmit radio waves (electromagnetic), but not sound.

25.2 Sound as a Longitudinal Mechanical Wave

Sound is a longitudinal mechanical wave. The medium's particles (air molecules, for instance) oscillate back and forth parallel to the direction of sound propagation, not perpendicular to it. This distinguishes sound from waves on a string or from light, which are transverse.

When a loudspeaker cone moves forward, it pushes the air molecules immediately in front of it together, creating a local region of higher pressure and density called a compression. When the cone moves backward, it pulls the air molecules apart, creating a region of lower pressure and density called a rarefaction. The speaker alternates rapidly between these two motions, creating an alternating sequence of compressions and rarefactions that travels outward as a sound wave.

Each individual air molecule does not travel with the sound — it oscillates back and forth around its original position, displaced only a tiny distance from equilibrium (typically much less than 1 mm, even for loud sounds). The pattern of compressions and rarefactions is what travels. This is exactly the wave/matter distinction from Chapter 24 applied to sound.

The Rule: Sound is longitudinal: medium particles oscillate parallel to the wave's direction of travel. The compressions and rarefactions are the wave — the medium particles merely oscillate and return to equilibrium.

25.3 Compression and Rarefaction

A compression is a region in a longitudinal wave where the medium's particles are pushed closer together than normal — a region of higher-than-normal pressure and density. A rarefaction is a region where particles are more spread out than normal — a region of lower-than-normal pressure and density.

In a sound wave in air, the pressure oscillates sinusoidally around atmospheric pressure. The amplitude of this pressure variation is typically extremely small compared to atmospheric pressure. A sound at the threshold of hearing (0 dB) has a pressure amplitude of only 2×10^{-5} Pa — about 2×10^{-10} times atmospheric pressure. A sound that is painfully loud (120 dB) has a pressure amplitude of about 20 Pa — still only about 0.02% of atmospheric pressure. The ear is extraordinarily sensitive.

The spatial wavelength of a sound wave is the distance between successive compressions (or successive rarefactions). At 343 m/s (speed of sound in air at 20°C) and 1000 Hz, the wavelength is 0.343 m. At 100 Hz, $\lambda = 3.43$ m. At 10 000 Hz, $\lambda = 0.0343$ m = 3.43 cm. These very different wavelengths explain why different audio frequencies interact so differently with rooms, instruments, and obstacles.

25.4 Speed of Sound in Different Media

The speed of sound in a medium depends on two competing factors: the stiffness (elasticity) of the medium, which speeds the wave up, and the density (inertia) of the medium, which slows it down. Stiffer, less dense media transmit sound faster.

$$v_{\text{sound}} = \sqrt{(B / \rho)} \quad (\text{for gases and liquids})$$

B = bulk modulus (a measure of the medium's resistance to compression); ρ = density. Stiffer and less dense $\rightarrow$ faster sound.

For air, the speed of sound also depends on temperature, because higher temperature increases molecular kinetic energy (which increases the effective stiffness):

$$v_{\text{air}} \approx 331 + 0.6T^{\circ}\text{C} \quad (\text{m/s})$$

Approximate speed of sound in air as a function of Celsius temperature. At 20°C: $v \approx 331 + 12 = 343$ m/s.

Medium	Speed of Sound (m/s)	Relative to Air
Air (0°C)	331	1.0× (reference)
Air (20°C)	343	1.04×
Water (20°C)	1480	4.3×
Seawater (20°C)	1522	4.4×
Wood (oak, along grain)	≈ 3800	: 11×
Concrete	≈ 3400	: 10×
Steel	≈ 5100	: 15×
Bone	≈ 3500	: 10×

Table 25.1 *Speed of sound in common media. Sound travels much faster in solids and liquids than in air.*

Sound travels faster in denser materials like water and steel because their greater bulk modulus (stiffness) outweighs the effect of their greater density. This is why you can hear a distant train by pressing your ear to the rail long before you hear it through the air. Sonar (Sound Navigation And Ranging) exploits the high speed and good propagation of sound in water to detect submarines, map the ocean floor, and locate schools of fish.

25.5 Frequency, Pitch, and Human Hearing

The frequency of a sound wave is the number of complete oscillation cycles per second, measured in hertz (Hz). Pitch is the subjective human perception of frequency. In general, higher frequency produces a higher perceived pitch. However, the relationship between frequency and pitch is not linear — pitch perception is logarithmic: doubling the frequency raises the pitch by one musical octave, regardless of the starting frequency.

The human auditory range spans from about 20 Hz (felt as much as heard) to about 20 000 Hz (a very high hiss). This range narrows with age; most adults over 40 cannot hear above 14 000–16 000 Hz. The ear is most sensitive in the range 1000–4000 Hz, which corresponds to the range of frequencies most important for speech intelligibility.

Frequency Range	Classification	Examples
Below 20 Hz	Infrasound	Earthquakes, large machinery, elephant calls
20 Hz – 20 kHz	Audible sound (human range)	Speech, music, environmental sounds
20 kHz – 1 GHz	Ultrasound	Medical imaging, sonar, industrial testing
Above 1 GHz	Hypersound	Specialized research applications

Table 25.2 *The spectrum of acoustic frequencies.*

Musical pitch is standardized internationally: concert A = 440 Hz. The twelve-note chromatic scale divides each octave (frequency ratio 2:1) into twelve equal steps (semitones), each with a frequency ratio of $2^{(1/12)} \approx 1.059$. This is equal temperament tuning, adopted universally in Western music since the 18th century.

Historical Note: The relationship between musical intervals and frequency ratios was understood by Pythagoras (c. 570–495 BC), who recognized that strings whose lengths

were in simple integer ratios (2:1, 3:2, 4:3) produced harmonious intervals. This discovery — that mathematics underlies music — was among the earliest examples of mathematical physics.

25.6 Amplitude, Intensity, and Loudness

The amplitude of a sound wave is the maximum pressure variation above and below atmospheric pressure. Intensity is the power per unit area of the wave, and it is proportional to the square of the amplitude, just as for all waves: $I \propto A^2$. The SI unit of intensity is watts per square meter (W/m^2).

Loudness is the subjective human perception of intensity. Like pitch, loudness perception is not linear in intensity — it is approximately logarithmic. An increase by a factor of 10 in intensity corresponds to roughly a doubling of perceived loudness (a very approximate rule, since loudness also depends on frequency).

The threshold of hearing, $I_0 = 1.0 \times 10^{-12} \text{ W/m}^2$, is the faintest sound an average human can detect at 1000 Hz. The threshold of pain is about 1.0 W/m^2 — a factor of 10^{12} larger. This enormous dynamic range is handled by the logarithmic decibel scale.

25.7 The Decibel Scale

The decibel (dB) is a logarithmic unit used to express sound intensity level. It compresses the vast range of audible intensities into a manageable numerical scale and better matches the logarithmic perception of the ear.

$$\beta = 10 \log_{10}(I / I_0) \quad (\text{decibels, dB})$$

Sound intensity level β in decibels. I = measured intensity; $I_0 = 10^{-12} \text{ W/m}^2$ (threshold of hearing reference). $\log_{10}$ is logarithm base 10.

Key properties of the decibel scale: the threshold of hearing is 0 dB ($I = I_0$). A 10 dB increase corresponds to a factor of 10 increase in intensity. A 3 dB increase corresponds to approximately doubling the intensity. Every 20 dB increase corresponds to a factor of 100 increase in intensity.

Sound	Intensity (W/m^2)	Level (dB)	Effect
Threshold of hearing	1.0×10^{-12}	0	—
Rustling leaves	1.0×10^{-11}	10	Barely perceptible

Whisper	1.0×10^{-10}	20	Very quiet
Library	1.0×10^{-7}	50	Quiet
Normal conversation	1.0×10^{-6}	60	Comfortable
Heavy traffic	1.0×10^{-4}	80	Noisy
Rock concert	1.0×10^{-1}	110	Risk of hearing damage
Jet engine at 30 m	1.0×10^2	140	Threshold of pain

Table 25.3 Representative sound intensity levels and their effects.

Common Mistake: Prolonged exposure to sounds above 85 dB causes permanent hearing damage. A factor-of-10 reduction in time of exposure only reduces cumulative damage linearly, while a 3 dB reduction in level halves the intensity and thus the rate of damage. Always use hearing protection near loud machinery, power tools, or amplified music.

Example — Decibel calculation A sound has intensity $I = 1.0 \times 10^{-8} \text{ W/m}^2$. Level: $\beta = 10 \log(10^{-8} / 10^{-12}) = 10 \log(10^4) = 10 \times 4 = 40 \text{ dB}$.

25.8 Reflection and Echoes

Sound reflects from surfaces following the same law of reflection that governs all waves: the angle of reflection equals the angle of incidence. When the reflected sound reaches a listener separately from the direct sound, it is perceived as an echo.

An echo is distinguishable from the original sound when the reflected sound arrives more than about 0.1 seconds after the direct sound — because the auditory system blends sounds arriving within 0.1 s into a single perception. At 343 m/s, a 0.1 s delay corresponds to a round-trip distance of 34.3 m, so an echo requires a reflecting surface at least 17 m away.

Reverberation is the persistence of sound in an enclosed space due to multiple reflections from walls, floor, and ceiling. Unlike an echo, reverberation is a blending of many closely-spaced reflections that prolongs the sound without producing distinct repeated events. The reverberation time (RT60) is the time for the sound intensity to drop by 60 dB after the source is turned off.

Reverberation time is critically important in architectural acoustics. Concert halls for orchestral music require relatively long reverberation times (1.8–2.2 s) that enhance

richness and blend. Speech auditoria and recording studios require shorter times (0.4–0.8 s) for clarity and intelligibility. Churches and cathedrals often have very long reverberation times (4–8 s or more), creating the reverberant quality associated with sacred choral music.

Sonar uses sound reflection to measure distances. A pulse of ultrasound is emitted, and the time for the echo to return is measured. Since $v = 1480 \text{ m/s}$ in water, and the sound travels there and back, $\text{distance} = v \times t / 2$. Medical ultrasound uses the same principle at frequencies of 2–15 MHz to image soft tissue with resolutions of less than 1 mm.

25.9 Standing Sound Waves

Just as standing transverse waves can form on a string, standing longitudinal waves can form in air columns inside tubes. When sound reflects back and forth between the ends of a tube, the incident and reflected waves superimpose. At specific resonant frequencies, this produces a stable standing wave pattern.

The boundary conditions at the ends of the tube determine the resonant frequencies. At a closed end, the air cannot move (the wall constrains it), creating a displacement node. At an open end, the air pressure is fixed at atmospheric pressure (the pressure wave meets the external air), creating a pressure node — equivalently, a displacement antinode.

For a tube open at both ends (like a flute or organ pipe open at the top), both ends are displacement antinodes. The resonant condition is that the tube length must be a whole number of half-wavelengths:

$$f_n = n \times v / (2L) \quad (n = 1, 2, 3, \dots) \quad [\text{open-open tube}]$$

Resonant frequencies for a tube open at both ends. Same formula as for a string fixed at both ends. The full harmonic series (all integers) is present.

For a tube closed at one end and open at the other (like a clarinet or closed organ pipe), one end is a displacement node and the other an antinode. Only odd harmonics are present:

$$f_n = n \times v / (4L) \quad (n = 1, 3, 5, \dots) \quad [\text{closed-open tube}]$$

Resonant frequencies for a tube closed at one end. Only odd harmonics appear. The fundamental frequency is half that of an open tube of the same length.

25.10 Resonance in Air Columns and Strings

Resonance in acoustics is the dramatic increase in vibration amplitude when a system is driven at one of its natural frequencies. For air columns and strings, the natural frequencies are the harmonic series derived in Section 25.9 and in Chapter 24.

A tuning fork held over a tube will cause the air column to resonate when the tube length is adjusted so that one of the column's natural frequencies matches the fork's frequency. This is a simple but precise method for measuring the speed of sound: measure the tube length L at resonance, note the frequency f of the fork, and apply $v = f\lambda = f \times 4L$ (for a closed-end tube at its fundamental).

Acoustic resonance shapes every musical instrument. The resonance of an instrument body amplifies specific frequencies from the vibrating source (string, reed, lips) and suppresses others. The combination of amplified frequencies — the instrument's resonance spectrum — determines its characteristic sound quality. This is why a violin and a viola playing the same note sound different: they have different bodies with different resonant characteristics.

Example — Resonance in a closed tube A tuning fork at 440 Hz causes a closed-end tube to resonate when $L = 0.195$ m. Speed of sound: $v = 4Lf = 4 \times 0.195 \times 440 = 343$ m/s. This is the standard speed of sound in air at 20°C, confirming the measurement.

Resonance in buildings and structures can be dangerous. Earthquakes produce seismic waves at frequencies of 0.1–10 Hz. Buildings of different heights have different natural frequencies — tall buildings resonate at lower frequencies than short ones. When the earthquake frequency matches the building's natural frequency, resonance amplifies the oscillation and can cause collapse. The 1985 Mexico City earthquake was particularly devastating partly because it produced waves near 2 Hz, matching the natural frequency of medium-rise buildings (7–15 stories) common in the city.

25.11 Harmonics and Overtones

The resonant frequencies of any vibrating system form a harmonic series. The fundamental (first harmonic) is the lowest resonant frequency. Higher resonant frequencies are called harmonics or overtones.

For a system with all harmonics present (open tube, string fixed at both ends), the n -th harmonic has frequency $f_n = n \times f_1$. The second harmonic is twice the fundamental; the third is three times; and so on. These frequencies form a simple integer ratio series.

For a system with only odd harmonics (closed-open tube, clarinet), the resonant frequencies are f_1 , $3f_1$, $5f_1$, ... The second resonance is at three times the fundamental, not twice it. This is why clarinets have a different tonal quality from flutes of similar size — the missing even harmonics give the clarinet its characteristic hollow, reedy quality.

Timbre (tone color) is the quality that distinguishes two instruments playing the same note at the same loudness. It is determined by the overtone spectrum: which harmonics are present and at what relative amplitudes. A piano, violin, and flute all playing concert A (440 Hz) produce the same fundamental frequency but radically different overtone spectra, which is why they sound so different. The ear performs a real-time Fourier analysis of the incoming sound, detecting the frequencies and amplitudes of each harmonic separately.

Harmonic	Frequency	Open/String Both-End	Closed-Open Tube
1st (fundamental)	f_1	$f_1 = v/2L$	$f_1 = v/4L$
2nd	$2f_1$	Present	Absent
3rd	$3f_1$	Present	Present (2nd resonance)
4th	$4f_1$	Present	Absent
5th	$5f_1$	Present	Present (3rd resonance)

Table 25.4 Harmonic series for open-open and closed-open tube configurations.

25.12 Musical Instruments and Sound Production

Musical instruments are carefully engineered acoustic resonators. They transform the energy of a player's input (bowing, blowing, striking) into sound by exciting and sustaining resonant oscillations at specific frequencies. Understanding the physics of instruments reveals why they are built as they are.

String instruments (violin, guitar, cello, piano) use a vibrating string as the source. The string's fundamental frequency is $f_1 = v_{\text{string}} / (2L)$, where $v_{\text{string}} = \sqrt{T/\mu}$ depends on string tension T and linear density μ . Increasing tension raises pitch (tuning). Shortening the vibrating length by pressing the string against the fingerboard also raises pitch (fretting or bowing position). Heavier strings vibrate at lower frequencies for the same tension (bass strings versus treble strings). The string alone produces very

little sound; the instrument body resonates and amplifies the string's vibration, radiating it efficiently into the air.

Wind instruments (flute, clarinet, trumpet, organ) use resonating air columns. The player's lips, reed, or embouchure provides the driving oscillation; the tube selects and amplifies specific resonant frequencies. Opening and closing tone holes (flute, clarinet) or valves (trumpet, French horn) changes the effective length of the air column, changing the resonant frequencies and thus the pitch. Brass instruments can also access different harmonics by changing lip tension and air pressure, a technique called overblowing.

Percussion instruments (drums, bells, xylophones, cymbals) produce sound through impact. Membranes, bars, and plates have complex resonant patterns that are not simple harmonic series. The irregular frequency spacing of percussion overtones gives them their non-pitched or metallic character. Tuned percussion (marimba, glockenspiel) is carefully shaped so that the first few resonant modes are in near-integer ratios, giving clear pitch.

25.13 Beats and Interference in Sound

When two sound waves of slightly different frequencies are superimposed, they alternate between constructive and destructive interference as their relative phase shifts cyclically. The result is a periodic variation in amplitude called beats. The beat frequency is the difference between the two source frequencies:

$$f_{\text{beat}} = |f_1 - f_2|$$

Beat frequency: the number of amplitude maxima per second when two nearby frequencies are superimposed. Beats are heard as a rhythmic pulsing or throbbing of the combined sound.

Beats are heard as a periodic rising and falling of loudness at the beat frequency. If two guitar strings are tuned to 440 Hz and 443 Hz, the combined sound will throb 3 times per second. Musicians use beats to tune instruments: when the beat frequency approaches zero, the two sources are converging on the same pitch. When beats disappear entirely, the instruments are in tune.

Beats also appear between a note and any of its harmonics, and between harmonics of different notes. This is partly responsible for the sense of consonance and dissonance in music: note combinations whose harmonics have few near-coincidences sound consonant (harmonious); combinations with many near-coincidences produce rapid beats perceived as rough or dissonant.

Constructive and destructive interference of sound are audible in many real

environments. In a large room with hard walls, standing at different positions relative to speakers can produce dramatically louder or quieter sound as path-length differences shift between constructive and destructive interference. Sound engineers exploit this to design speaker arrangements that produce uniform coverage; they avoid positions of systematic destructive interference called nulls.

25.14 The Doppler Effect (Introductory)

The Doppler effect is the change in observed frequency of a wave when the source or the observer is in motion relative to the medium. When source and observer approach each other, the observed frequency is higher than the emitted frequency. When they move apart, the observed frequency is lower.

The physical explanation is straightforward for a moving source. If a car honks its horn while moving toward you, each successive wavefront is emitted slightly closer to you than the previous one (because the car has moved forward). The wavefronts are compressed together, shortening the wavelength and increasing the frequency you observe. When the car passes and moves away, the wavefronts are stretched, lengthening the wavelength and lowering the observed frequency. This is the familiar drop in pitch as a vehicle passes.

$$f_{\text{observed}} = f_{\text{source}} \times (v \pm v_{\text{observer}}) / (v \mp v_{\text{source}})$$

Doppler formula. v = wave speed; v_{observer} and v_{source} are speeds relative to the medium. Upper signs for approach; lower signs for recession. Use the sign that gives higher frequency for approach.

Doppler shift is used extensively in technology. Police radar and speed cameras use the Doppler shift of microwave or laser pulses to measure vehicle speed. Medical Doppler ultrasound measures blood flow velocity inside the body by detecting the frequency shift of ultrasound reflected from moving blood cells. Astronomical redshift — the Doppler shift of light from distant galaxies — provided the first evidence that the universe is expanding. The Doppler shift of stars in binary systems reveals orbital velocities and masses.

Historical Note: Christian Doppler (1803–1853) described the effect in 1842 in a paper on the colored light of double stars. The effect was confirmed acoustically the following year in a colorful experiment: a locomotive was used to carry a group of horn players past a group of musicians with perfect pitch, who listened and reported the pitch shift. This was science conducted in the grandest Victorian tradition.

Bridge Note *The Doppler formula given here applies when all speeds are much less than*

the wave speed. For sound near the speed of sound, more complex shock-wave physics applies. For light, the relativistic Doppler formula (derived from special relativity) is required, and is the basis for measuring cosmological expansion.

25.15 Common Mistakes in Acoustics Analysis

- Treating sound as a transverse wave. Sound is longitudinal. The oscillations of air molecules are back-and-forth along the direction of propagation, not perpendicular to it.
- Using the wrong formula for closed vs. open tubes. An open-open tube supports all harmonics with $f_n = nv/(2L)$. A closed-open tube supports only odd harmonics with $f_n = nv/(4L)$. These are different formulas; using the wrong one gives frequencies off by a factor of 2 or gives harmonics that should not be present.
- Forgetting that the decibel scale is logarithmic. Adding two sources of equal intensity does not double the decibel level; it increases it by 3 dB. Adding ten equal sources increases the level by 10 dB. Doubling the decibel value does not double the intensity — it increases the intensity by a factor of $10^{(dB_increase/10)}$.
- Confusing loudness (perception) with intensity (physics). A doubling of perceived loudness corresponds to roughly a 10 dB (factor-of-10 intensity) increase for mid-range frequencies. They are not proportional.
- Applying the beat formula to widely different frequencies. Beats are perceptible only when the two frequencies are close (a few Hz difference). The formula $f_{beat} = |f_1 - f_2|$ gives the correct result mathematically, but for large differences the result is not perceived as a beat frequency.
- Getting the Doppler signs wrong. Remember: approach → higher frequency; recession → lower frequency. When the observer moves toward the source, use $f = f_s(v + v_o)/v$. When the source moves toward the observer, use $f = f_s \times v/(v - v_s)$. Never subtract in the wrong place.
- Confusing echo (distinct reflected sound) with reverberation (blending of many reflections). Echoes require reflecting surfaces at least 17 m away. Reverberation occurs in any enclosed space.

Chapter Summary

Sound is a longitudinal mechanical wave — compressions and rarefactions propagating

through a material medium. It cannot travel through a vacuum.

- Speed of sound depends on medium stiffness and density: $v \approx 331 + 0.6T^{\circ}\text{C}$ in air. Sound travels $\sim 4\times$ faster in water and $\sim 15\times$ faster in steel than in air.
- Frequency determines pitch; amplitude determines intensity and loudness. Human hearing spans 20 Hz to 20 kHz.
- Intensity I is measured in W/m^2 . Sound level β in decibels: $\beta = 10 \log(I/I_0)$. A 10 dB increase = $10\times$ intensity. A 3 dB increase = $2\times$ intensity.
- Standing waves in open-open tubes: all harmonics, $f_n = nv/(2L)$. In closed-open tubes: odd harmonics only, $f_n = nv/(4L)$.
- Timbre is determined by the overtone spectrum — which harmonics are present and at what relative amplitudes.
- Beats arise from superposition of nearly equal frequencies: $f_{\text{beat}} = |f_1 - f_2|$. Used for instrument tuning.
- Doppler effect: moving source or observer changes observed frequency. Approach $\rightarrow$ higher f ; recession $\rightarrow$ lower f . Used in radar, medical ultrasound, and astronomy.

Key Terms

sound: A longitudinal mechanical wave consisting of alternating compressions and rarefactions propagating through a material medium.

compression: A region of a longitudinal wave where particles are closer together than normal, producing higher-than-ambient pressure and density.

rarefaction: A region of a longitudinal wave where particles are farther apart than normal, producing lower-than-ambient pressure and density.

infrasound: Sound below 20 Hz, below the human auditory range; produced by earthquakes, large machinery, and some animals.

ultrasound: Sound above 20 kHz, above the human auditory range; used in medical imaging, sonar, and industrial applications.

pitch: The subjective human perception of sound frequency; higher frequency produces higher pitch; perception is approximately logarithmic.

intensity (I): Power delivered per unit area perpendicular to wave propagation; $I = P/A$; measured in W/m^2 .

threshold of hearing (I_0): The minimum intensity audible to the average human at

1000 Hz: $I_0 = 1.0 \times 10^{-12} \text{ W/m}^2$.

decibel (dB): Logarithmic unit of sound intensity level; $\beta = 10 \log(I/I_0)$. A 10 dB increase corresponds to a factor-of-10 increase in intensity.

echo: A reflected sound that arrives at the observer distinctly after the direct sound, when the reflecting surface is at least 17 m away.

reverberation: The persistence of sound in an enclosed space due to multiple reflections; characterized by the RT60 reverberation time.

sonar: Sound Navigation And Ranging; uses sound pulses and echo timing to measure distances in water.

standing sound wave: A stationary wave pattern in an air column formed by superposition of incident and reflected longitudinal waves at resonant frequencies.

fundamental frequency: The lowest resonant frequency of a standing wave system; all higher harmonics are integer multiples of the fundamental.

harmonic: A resonant frequency of a vibrating system; the n -th harmonic has frequency n times the fundamental.

timbre: The tone quality or tone color of a sound; determined by the overtone spectrum (which harmonics are present and at what amplitudes).

beats: Periodic variation in amplitude produced by the superposition of two waves of slightly different frequencies; beat frequency $= |f_1 - f_2|$.

Doppler effect: The change in observed wave frequency when source or observer moves relative to the medium; approach increases frequency, recession decreases it.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Explain why sound cannot travel through a vacuum. What does this tell us about the mechanism of sound propagation?
2. Describe the motion of an air molecule as a sound wave passes. Is it transverse or longitudinal motion? How does the molecule's position change over one complete cycle?
3. A submarine sonar system emits a pulse and receives the echo 0.80 s later. Explain what the crew knows from this information. What additional information would they need to locate the object precisely?

4. Two concerts are advertised: one at 80 dB and one at 90 dB. How many times more intense is the 90 dB concert? By what factor must you increase distance from a speaker to reduce the level by 10 dB (assume inverse-square spreading)?
 5. Explain the difference between an open-open air column and a closed-open air column. Which one produces odd harmonics only? Why does the closed end create a displacement node?
 6. What is timbre, and what physical property of a sound wave determines it? Why does a violin sound different from a flute playing the same note at the same loudness?
 7. Explain the physics of beats. Why do musicians use beats to tune instruments? What does it mean when the beats disappear?
 8. A police radar gun uses the Doppler effect to measure vehicle speed. Explain the principle. Why does the gun measure the frequency shift of the reflected microwave rather than sending a continuous beam and simply timing the travel?
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Sound Wave Problems

9. Find the speed of sound in air at (a) 0°C, (b) 20°C, and (c) 40°C using $v \approx 331 + 0.6T^\circ\text{C}$.
 10. A thunder clap is heard 4.5 s after the lightning flash. How far away is the lightning? (Use $v = 343 \text{ m/s}$.)
 11. A sonar pulse returns after 0.300 s. How far away is the reflecting object? (Speed of sound in seawater = 1522 m/s.)
 12. Medical ultrasound at 5.0 MHz travels through soft tissue at 1540 m/s. Find the wavelength.
 13. A sound has intensity $4.0 \times 10^{-6} \text{ W/m}^2$. Calculate its intensity level in dB.
 14. A sound level meter reads 75 dB. Find the intensity in W/m^2 .
 15. A point source emits sound at 50 W. Find the intensity at (a) 1.0 m, (b) 5.0 m, and (c) 10 m. Find the sound level at each distance.
 16. Two workers near a machine each experience 88 dB. If they stand side by side (combining their individual contributions as if each adds an independent 88 dB source), by how many dB does the level increase?
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Resonance Exercises

17. An open-open tube is 0.400 m long. Speed of sound = 340 m/s. Find the frequencies of the first three harmonics.
 18. A closed-open tube (clarinet approximation) is 0.600 m long ($v = 343$ m/s). Find the fundamental frequency and the first two overtones (3rd and 5th harmonics).
 19. A tuning fork of 512 Hz is held over a closed-open tube that can be lengthened by sliding a plunger. At what tube length does the first resonance occur? The second?
 20. An organ pipe open at both ends has a fundamental of 64 Hz at 20°C. How long is the pipe? If the temperature drops to 0°C, what is the new fundamental frequency?
 21. A guitar string has length 0.650 m, mass per unit length 4.0×10^{-3} kg/m, and is under tension 80 N. Find (a) the wave speed in the string, (b) the fundamental frequency, and (c) the frequencies of the 2nd, 3rd, and 4th harmonics.
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Harmonics Problems

22. A string fixed at both ends has fundamental frequency 120 Hz. List the frequencies of the first five harmonics. Which of these harmonics would also be present in a closed-open tube of the same fundamental?
 23. A flute (open-open) and a clarinet (closed-open) have the same length, 0.580 m. Compare their fundamental frequencies at $v = 343$ m/s. Explain why the clarinet sounds an octave lower than a flute of the same length.
 24. Two speakers emit 440 Hz and 444 Hz. (a) What beat frequency do you hear? (b) After the beats disappear, the 444 Hz speaker is retuned to 437 Hz. What beat frequency is now heard?
 25. A piano string and a tuning fork at 440 Hz produce 3 beats per second. The string could be at 437 Hz or 443 Hz. When the string tension is slightly increased, the beat frequency increases to 5 beats per second. What is the actual frequency of the string? Explain your reasoning.
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Applied Acoustics Analysis

26. A concert hall has a reverberation time of 2.0 s at 500 Hz. An echo room for special effects has $RT_{60} = 8.0$ s. A recording studio has $RT_{60} = 0.3$ s. Explain which environment is appropriate for (a) orchestral performance, (b) spoken-word podcasting, (c) digital reverb simulation.
27. A bat emits 45 kHz ultrasound and detects the echo 3.6×10^{-3} s later. How far away is the insect? ($v = 343$ m/s.) If the insect is flying toward the bat at 5.0 m/s, what is the frequency of the echo received by the bat (both source and observer are moving — apply Doppler twice)?
28. A fire truck approaches you at 25 m/s sounding its siren at 880 Hz. ($v = 343$ m/s.) (a) What frequency do you hear as it approaches? (b) As it recedes? (c) What is the beat frequency you hear as the truck is momentarily alongside you (when you momentarily hear both frequencies superimposed in the transition)?
29. An ultrasound probe at 4.0 MHz detects blood flow in an artery. The probe is angled at 60° to the blood vessel. The received echo is 4.0008 MHz. Find the component of blood velocity along the probe axis. (Use the Doppler formula with $v_{\text{blood}} \ll v_{\text{sound}} = 1540$ m/s, so the frequency shift $\Delta f \approx f_0 \times 2v_{\text{blood}} \cos\theta / v_{\text{sound}}$.)

Chapter 25 Checkpoint

Before moving on to the next chapter, confirm you can do each of the following.

I can...	See Section
State what sound is and why it cannot travel through a vacuum	25.1
Explain how compressions and rarefactions create a longitudinal wave	25.2–25.3
Calculate the speed of sound in air at a given temperature	25.4
Define frequency, pitch, and the human auditory range	25.5
Define intensity and use the decibel formula $\beta = 10 \log(I/I_0)$	25.6–25.7
Explain echo, reverberation, and how sonar works	25.8
Calculate resonant frequencies for open-open and closed-open air columns	25.9

Explain how resonance is used to measure the speed of sound	25.10
Distinguish the harmonic series for open-open vs. closed-open systems	25.11
Explain how string tension and length affect a string's pitch	25.12
Calculate beat frequency and explain how it is used for tuning	25.13
Describe the Doppler effect and apply the Doppler formula	25.14
Identify and avoid common errors in acoustics problems	25.15

Continue to Chapter 26: Light and the Electromagnetic Spectrum →